MILITARY SPECIFICATION

CUTTER, POWDER ACTUATED, REEFING LINE, XM23 ASSEMBLY

1. SCOPE

1.1 This specification covers Cutter, Powder Actuated, Reefing Line, XM23, Assembly loaded with Cartridge, Powder Actuated Cutter, M-129 for use in connection with cargo parachute controlled airdrops.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons, General Specification for

MIL-W-5625 — Webbing, Textile, Nylon, Tubular

MIL-I-45208 — Inspection Requirements, General Specification for

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-109 — Quality Assurance Terms and Definitions

MIL-STD-414 — Sampling Procedures and Tables for Inspection by Variables for Percent Defective

DRAWINGS

MUNITIONS COMMAND

D10524020 — Cutter, Powder Actuated, Reefing Line, XM23, Assembly

C8864326 — Box, Packing, Ammunition, for Cutter, Powder Actuated Reefing Line, M9, XM10, XM11, XM-12, M13 or XM23 with Cartridge

C8864327 — Carton, Packing, Ammunition, for Cutter, Powder Actuated, Reefing Line, M9, XM10, XM11, XM12, M13 or XM-23, with Cartridge

IEL-10524020 — Inspection Equipment List for Cutter, Powder Actuated, Reefing Line, XM-23, Assembly

FSC 1375

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(Copies of specifications, standards and drawings required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

INTERSTATE COMMERCE COMMISSION — 49 CFR 71–78

The Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles. (The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations, available from the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders for the above publication should cite "49 CFR 71–78".)

3. REQUIREMENTS

3.1 General. Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Functional.

3.2.1 The cutter shall not fail to fire.

3.2.2 The cutter shall function without mechanical failure (see 6.1).

3.2.3 *Function.* The cutter, loaded with an accepted M–129 Cartridge, temperature conditioned, as required in table I, shall completely sever two (2) lines of ½ inch, 1000 pound tubular nylon webbing fabricated in accordance with Specification MIL–W–5625, within the delay times specified in table I.

TABLE I.

Temperature	Delay time, seconds (see 6.2)
—65° to —70° F.	2.4 max.
65° to 75° F.	1.8 min. to 2.2 max.
160° to 165° F.	1.6 min.

3.3 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.3.1 *Metal defects.* All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.3.2 *Burr.* No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.3.3 *Foreign matter.* No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Quality assurance terms and definitions.* Reference shall be made to Standard MIL–STD–109 to define quality assurance terms used.

4.1.2 *Inspection.* Inspection shall be in accordance with Specifications MIL–A–2550 and MIL–I–45208, except as specified herein.

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4.2 Inspection provisions.

4.2.1 *Lot.*

4.2.1.1 *Presentation of lots.* The product shall be presented in accordance with Standard MIL-STD-105 or MIL-STD-414, as applicable.

4.2.1.2 *Size of lot.* A lot shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogenous process and loaded with a single lot of cartridges.

4.2.2 *Examination.* One hundred percent examination shall be performed for all critical defects. Examination for major or minor defects shall be performed on a defect basis in accordance with the classification of defects, using the sampling plan and criteria in tables II and III. For an individual defect (single characteristic), the criteria in tabl II shall be used. For total defects within a major or minor class, the criteria in table III shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as bur and foreign matter.

TABLE II.

Lot size	Sample size	Major		Minor	
		Accept	Reject	Accept	Reject
1 — 1300	110	1	2	2	3
1301 — 3200	150	2	3	3	4
3201 — 8000	225	3	4	4	5
8001 — 22000	300	4	5	5	6

TABLE III. *Total allowable defects per sample size*

Sample sizes for major defects	110	150	225	300
No. of Defects in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over 43	6	8	11	14
Sample sizes for minor defects	**110**	**150**	**225**	**300**
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	3	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

4.2.2.1 *Cartridge, M-129, Drawing D10520297.*

Defect	Method of inspection
Critical	
1. Mixed cartridges	Visual
2. Metal defective	Visual
3. O-ring missing	Visual
4. Marking missing, incorrect or illegible	Visual

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Defect	Method of inspection
Major	
101. O-ring split, cut or cracked	Visual
102. Foreign matter	Visual
Minor	
201. Case scratched or dented	Visual

4.2.2.2 *Cutter, Powder Actuated, Reefing Line, XM23, Assembly, Drawing D10524020.*

Defect	Method of inspection
Critical	
1. Firing pin not cocked	X-Ray/Manual [1]
2. Any part missing	X-Ray/Visual [1]
Major	
101. Front of anvil not flush or slightly below reefing line holes	Visual
102. Metal defective	Visual
103. Security of assembly	Manual
Minor	
201. Length from anvil end to cable eyelet center, max.	Measure
202. Radii or chamfers, missing or incorrect	Visual/Measure
203. Reefing line holes obstructed	Visual
204. Marking missing, incorrect or illegible	Visual
205. Surface finish improper	Visual
206. Protective finish inadequate or defective	Visual
207. Bur	Visual
208. Foreign matter	Visual

[1] The cutter shall be x-rayed with the long axis of the reefing line cutter perpendicular to the rays of the x-ray machine.

4.2.2.3 *Carton, Packing, Ammunition, for Cutter, Powder Actuated, Reefing Line M9, XM10, XM11, XM12, M13, or XM23, with Cartridge, Drawing C8864327.*

Defect	Method of inspection
Critical	
None defined.	
Major	
None defined.	
Minor	
201. Quantity of contents incorrect	Visual
202. Contents loose	Manual/Visual
203. Separator or filler missing	Visual
204. Cut or damage through impregnated carton	Visual
205. Marking missing, incorrect or illegible	Visual
206. Carton sealing defective, improper or missing	Visual

4.2.2.4 *Box, Packing, Ammunition, for Cutter, Powder Actuated, Reefing Line, M9, XM10, XM11, XM12, M13, or XM23, with Cartridge, Drawing C8864326.*

Defect	Method of inspection
Critical	
None defined.	
Major	
None defined.	
Minor	
201. Quantity of contents incorrect	Visual
202. Contents loose	Manual/Visual

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Defect	Method of inspection
203. Contents of box exposed	Visual
204. Box damaged	Visual
205. Strapping missing, loose, broken or improperly applied	Visual
206. Marking missing, incorrect or illegible	Visual
207. Car seal missing, unsealed or improperly positioned	Visual

4.2.3 *Inspection testing.* The following tests shall be performed in accordance with the provisions of 4.3, except as provided in Specification MIL–I–45208.

4.2.3.1 *Functional.* The following tests shall be performed by the Government:

4.2.3.1.1 *Ballistics.* Twenty-one assembled reefing line cutters of each lot shall be grouped, temperature conditioned as specified in Table IV and 4.3.1.1.1 and then tested in accordance with 4.3.1.1. The variables sampling plan of Standard MIL–STD–414, example C–2 or C–3, as applicable, for the AQL listed in table IV shall be used for each group. Failure of any reefing line cutter to comply with the requirements of 3.2, or failure of a group to meet the criteria of the variables sampling plan shall be cause for rejection of the lot.

TABLE IV. *Sample size*

Temperature	Production lot	AQL
—65° to —70° F.	7	0.65%
65° to 75° F.	7	0.65%
160° to 165° F.	7	0.65%

4.2.4 *Inspection equipment.* The examination and tests shall be made using the equipment listed on IEL–10524020, except as provided in Specification MIL–I–45208.

4.3 Test methods and procedures.

4.3.1 *Functional.*

4.3.1.1 *Ballistics.* The test sample shall be rigidly supported in the fixture. The sample shall be tested within 5 minutes after removal from the conditioning chamber. When this time limit is exceeded, the sample shall be reconditioned at the original temperature.

4.3.1.1.1 *Temperature conditioning.* A standard XM23 Reefing Line Cutter, loaded with a dummy M–129 Cartridge containing a temperature sensitive element, shall be placed in the conditioning chamber at the same time as the sample. The conditioning time shall be a minimum of 1-hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking. Level A — Packing and marking shall be in accordance with Drawings C8864326 and C8864327.

5.2 Delivery. Delivery shall be made in accordance with Interstate Commerce Commission Regulations cited in 2.2.

5.3 Data cards. Data card information shall be as specified in Specification MIL–A–2550.

6. NOTES

6.1 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.2 "Delay time" is defined as the time elapsed between initiation of the cable withdrawal and complete severing of the reefing lines.

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